Week4 tut

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2024-02-02

Q1

i

```
Book1 = read.csv(file = "Book1.csv")
Book2 = read.csv(file = "Book2.csv")
```

ii

```
# Merge the datasets based on "location_id"
m_d1 = right_join(Book1, Book2, by="location_id")
```

iii

```
m_d2 = anti_join(Book2, Book1, by ="location_id")
```

iv

```
m_d1 %>% mutate(new_p_code = str_sub(LOCATION_POSTAL_CODE, start = 1, end = 2)) %>% grou
p_by(new_p_code) %>% summarize(new_var = sum(UNOCCUPIED_ROOMS, na.rm = T))
```

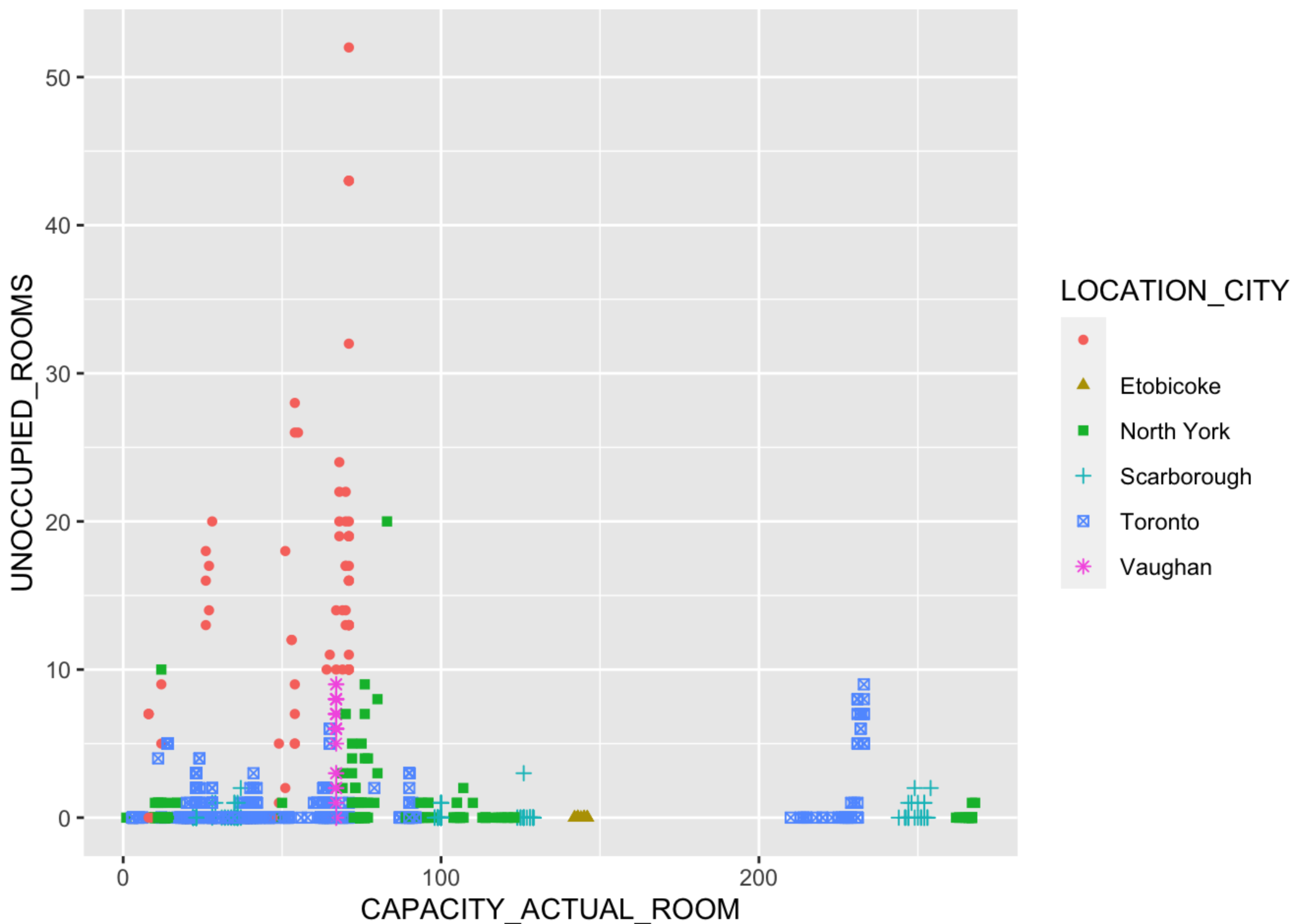
```
## # A tibble: 10 × 2
##    new_p_code new_var
##    <chr>        <int>
##  1 ""            1027
##  2 "L4"           155
##  3 "M1"            32
##  4 "M2"            11
##  5 "M3"           123
##  6 "M4"            26
##  7 "M5"           476
##  8 "M6"             2
##  9 "M8"             0
## 10 "M9"             0
```

Q2

i

```
ggplot(m_d1, aes(x = CAPACITY_ACTUAL_ROOM, y = UNOCCUPIED_ROOMS, color = LOCATION_CITY,
shape = LOCATION_CITY)) + geom_point()
```

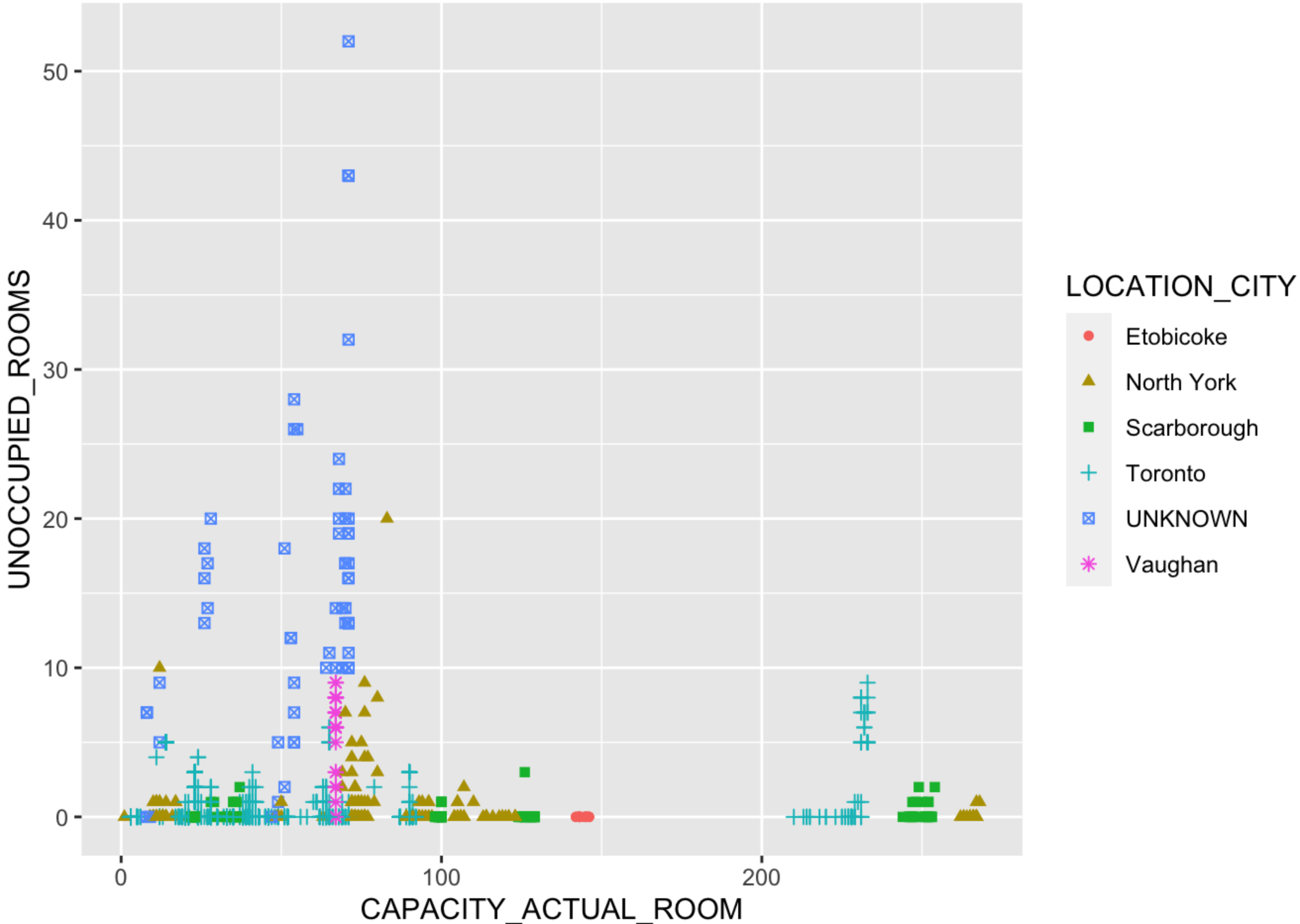
```
## Warning: Removed 2711 rows containing missing values (`geom_point()`).
```



ii

```
m_d1 = m_d1 %>% mutate( LOCATION_CITY = case_when(LOCATION_CITY == "" ~ "UNKNOWN",
                                                  TRUE ~ LOCATION_CITY))
ggplot(m_d1, aes(x = CAPACITY_ACTUAL_ROOM, y = UNOCCUPIED_ROOMS, color = LOCATION_CITY,
shape = LOCATION_CITY)) + geom_point()
```

```
## Warning: Removed 2711 rows containing missing values (`geom_point()`).
```


UNOCCUPIED_ROOMS
50
40
30
20
10
0
0
100
200
CAPACITY_ACTUAL_ROOM
LOCATION_CITY
Etobicoke
North York
Scarborough
Toronto
UNKNOWN
Vaughan